CHRISTOPHER MADARIAGA

Software Engineer | Data Scientist | Technology Leader

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PROFILE

Software engineer, data scientist, and technology leader with 15+ years of experience building enterprise software, data platforms, and business intelligence solutions, progressing from developer to Director of Enterprise Applications, where I lead programming, data science, and systems integration teams. Currently pursuing a Master of Computer Science and AI at Boston University to deepen foundations in systems programming, statistics, machine learning, and AI. Founder of Phantom Analytics LLC and co-founder of Future Explorers Academy, focused on applying data and AI to measurable business outcomes.

TECHNICAL SKILLS

Languages: Python, SQL, C, Java, C# / .NET

Data Science: NumPy, pandas, Jupyter, statistical inference, hypothesis testing, experimental design

Data & BI: Data warehousing, ETL (SSIS, Talend), Power BI, SSRS, Crystal Reports, OneStream

Enterprise Platforms: SAP, Oracle, NetSuite, Dynamics 365, SharePoint, Power Platform

Engineering & Tools: Git / GitHub, GitHub Codespaces, VS Code, Flask, SQLite, DevOps practices, AI / LLM tooling

EDUCATION

Boston University *Currently Enrolled*

Master of Computer Science and Artificial Intelligence (in progress)

- Coursework: systems programming in C (operating systems, file systems, networking), Java and JVM internals, statistics and data science in Python, experimental design, technical decision-making.

National Polytechnic Institute (IPN), Mexico City

Bachelor of Science, Computer Systems Engineering

ACADEMIC & TECHNICAL PROJECTS

- **Networked Connect 4 (C, UDP):** Built a client/server game with a custom message protocol, direction-based win detection, and save/load crash recovery; passed the full automated test suite.
- **Statistical Analysis in Python (NumPy, Jupyter):** Implemented z-scores, outlier detection, and paired t-tests, and designed controlled experiments with testable hypotheses.
- **School Registration Web App (Flask, SQLite):** Developed a registration system with field validation, password-protected admin view, and CSV export.

PROFESSIONAL EXPERIENCE

QUANEX BUILDING PRODUCTS | Houston, TX *2018 – Present*

Director of Enterprise Applications *2024 – Present*

- Lead teams of programmers, data scientists, and systems integration specialists delivering enterprise applications, analytics, and AI solutions.
- Own integration across enterprise systems, connecting ERP, BI, and operational platforms into a unified data landscape.
- Established the initial roadmap for the Enterprise Digital Data Hub, the foundation for future AI-driven capabilities.
- Lead AI research and evaluation focused on manufacturing productivity, analytics, automation, and software development.
- Partner with executive leadership to align technology investments and digital transformation with business strategy.

Applications Development Manager *2023 – 2024*

- Established enterprise development standards, architecture guidelines, governance, and DevOps best practices.

- Led global application development for Finance, Operations, Manufacturing, and Corporate functions, including the CAPEX Tracker.
- Played a key technology leadership role in the Tyman acquisition and systems integration.

Senior Programmer & Analyst *2020 – 2023*

- Designed and delivered enterprise applications in .NET, SharePoint, and Power Platform for Finance, Operations, HR, and Manufacturing.
- Built the Global Expenditure Tracker, AP Invoice Tracker, and Quanex Foundation Charity Application.
- Implemented manufacturing-floor applications for labeling, shipping, and packaging.

Business Analyst / SharePoint Administrator / Developer *2018 – 2020*

- Led development of QNET, the enterprise intranet supporting global collaboration.
- Consolidated multiple contract management systems into a unified enterprise solution.

ENTREPRENEURSHIP

Phantom Analytics LLC | Founder

- Founded a technology company delivering IT services, software development, analytics, and AI-powered solutions.

Future Explorers Academy | Co-Founder & Owner

- Co-founded and grew an early-education organization to nearly 200 students and families.
- Designed and developed a custom platform supporting parents, teachers, and administrators.

CERTIFICATIONS

Microsoft Certified Technology Specialist (MCTS): MOSS 2007 Application Development, Office SharePoint Server, SharePoint 2010 Application Development